

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

09

Q1**A****A. 14**

Choice A is correct. It's given that Julissa already has 86 hours of flight time. Let x represent the number of additional hours of flight time Julissa needs to get her private pilot certification.

After completing x hours of flight time, Julissa will have completed a total of $86 + x$ hours of flight time. It's given that Julissa needs at least 100 hours of flight time to get her private pilot certification. Therefore, $86 + x \geq 100$. Subtracting 86 from both sides of this inequality yields $x \geq 14$. Thus, 14 is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of hours of flight time Julissa already has, rather than the minimum number of additional hours of flight time Julissa needs.

Choice D is incorrect. This is the number of hours of flight time Julissa will have if she completes 100 more hours of flight time, rather than the minimum number of additional hours of flight time Julissa needs.

Q2**B**

B. $y = 7x + 5$

Choice B is correct. The equation of a line in the xy -plane can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is its y -intercept. It's given that the line passes through the point $(0, 5)$. Therefore, $b = 5$. It's also given that the line is parallel to the graph of $y = 7x + 4$, which means the line has the same slope as the graph of $y = 7x + 4$. The slope of the graph of $y = 7x + 4$ is 7. Therefore, $m = 7$. Substituting 7 for m and 5 for b in the equation $y = mx + b$ yields $y = 7x + 5$.

Choice A is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$, and has a slope of 5, not 7.

Choice C is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. The graph of this equation passes through the point $(0, 7)$, not $(0, 5)$, and has a slope of 5, not 7.

Q3**40**

The correct answer is 40. Subtracting 5 from both sides of the given equation yields $4x = 160$. Dividing both sides of this equation by 4 yields $x = 40$. Therefore, the solution to the given equation is 40.

Q4**A****A. \$2.00**

Choice A is correct. The cost of each additional mile traveled is represented by the slope of the given line. The slope of the line can be calculated by identifying two points on the line and then calculating the ratio of the change in y to the change in x between the two points. Using the points $(1,5)$ and $(2,7)$, the slope is equal to $\frac{7-5}{2-1}$, or 2. Therefore, the cost for each additional mile traveled of the cab ride is \$2.00.

Choice B is incorrect and may result from calculating the slope of the line that passes through the points $(5,13)$ and $(0,0)$. However, $(0,0)$ does not lie on the line shown. Choice C is incorrect. This is the y -coordinate of the y -intercept of the graph and represents the flat fee for a cab ride before the charge for any miles traveled is added. Choice D is incorrect. This value represents the total cost of a 1-mile cab ride.

Q5**B****B. 2, 2**

Choice B is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 2$ and $y = 2$, or at the point 2, 2. Therefore, the solution x, y to the system is 2, 2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**B**

B. $1,510 \leq d \leq 4,130$

Choice B is correct. It's given that during spring migration, a dragonfly traveled a minimum of 1,510 miles and a maximum of 4,130 miles between stopover locations. It's also given that d represents a possible distance, in miles, this dragonfly traveled between stopover locations. It follows that the inequality $1,510 \leq d \leq 4,130$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which a dragonfly traveled a maximum of 1,510 miles between stopover locations.

Choice C is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of 4,130 miles between stopover locations.

Choice D is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of 4,310 miles and a maximum of 5,640 miles between stopover locations.

Q7**A**

A. 5

Choice A is correct. Let x represent the number of rabbit snails that Naomi bought. It's given that each rabbit snail costs \$ 8. Therefore, the total cost, in dollars, of the rabbit snails that Naomi bought can be represented by the expression $8x$. It's also given that each nerite snail costs \$ 6, and that Naomi bought 2 nerite snails. Therefore, the total cost, in dollars, of the nerite snails that Naomi bought is 62, or 12. Since Naomi bought both the rabbit snails and the nerite snails for a total of \$ 52, the equation $8x + 12 = 52$ can be used to represent the situation. Subtracting 12 from both sides of this equation yields $8x = 40$. Dividing both sides of this equation by 8 yields $x = 5$. Therefore, Naomi bought 5 rabbit snails.

Choice B is incorrect. This is the total cost, in dollars, of the nerite snails that Naomi bought, not the number of rabbit snails.

Choice C is incorrect. This is the cost, in dollars, of one rabbit snail and one nerite snail, not the number of rabbit snails that Naomi bought.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**B****B.** 130

Choice B is correct. It's given that the film club has 90 members on the first day of a semester, and 10 new members join the film club each day after the first day of the semester. This means that after 4 days, 4×10 , or 40, new members will have joined the club. Adding 40 members to the original 90 club members yields 130 members. Thus, the film club will have 130 total members 4 days after the first day of the semester.

Choice A is incorrect. This is the number of members that will have joined the film club 4 days after the first day of the semester if 100 new members, not 10, join the film club each day.

Choice C is incorrect. This is the number of members the film club will have 4 days after the first day of the semester if 1 new member, not 10, joins the film club each day.

Choice D is incorrect. This is the number of members the film club has on the first day of the semester.

Q9**D****D.** 740

Choice D is correct. It's given that at the beginning of the 1st week of the year there was \$ 600 in a savings account and Gabriella deposits \$ 35 in that savings account at the end of each week. Therefore, the amount of money, in dollars, in the savings account at the end of the 4th week of that year is $600 + 435$, or 740.

Choice A is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 4th week if Gabriella withdraws, rather than deposits, \$ 35 at the end of each week.

Choice B is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 1st week, not the 4th week.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10

B

B. $(2, 4)$

Choice B is correct. A solution to a system of equations must be the solution to each equation in the system. It follows that if x, y is a solution to the system, the point x, y lies on the graph in the xy -plane of each equation in the system. The point that lies on each graph of the system of linear equations shown is their intersection point $2, 4$. Therefore, the solution to the system is $2, 4$.

Choice A is incorrect. The point $0, 5$ lies on one, but not both, of the graphs of the linear equations shown.

Choice C is incorrect. The point $5, 10$ lies on one, but not both, of the graphs of the linear equations shown.

Choice D is incorrect. The point $10, 0$ lies on one, but not both, of the graphs of the linear equations shown.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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